

David Kanter Eivin

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EDUCATION

Michael G. DeGroote School of Medicine, McMaster University

Niagara Regional Campus (NRC)

St. Catharines, ON

Doctor of Medicine (in progress)

Aug 2023 — May 2026

- Completed Area of Focused Professional Development in Research & Scholarship.

McMaster University

Hamilton, ON

Bachelors of Arts & Science

Sep 2020 — May 2023

- > 11/12 cGPA with courses in Computer Science, Life Sciences, Calculus (with proofs), Statistics, Scientific Computing, Machine Learning, and Medical Humanities.

CLINICAL ELECTIVES

Internal Medicine Experiences

Internal Medicine TtR - 2 weeks	McMaster University	Niagara Health - St. Catharine's Site	Feb 2026
GIM CTU Elective - 2 weeks	McMaster University	Hamilton General Hospital	Nov 2025
GIM CTU Elective - 2 weeks	McMaster University	Niagara Health - St. Catharine's Site	Jun 2025
Internal Medicine Core - 4 weeks	McMaster University	Niagara Health - St. Catharine's Site	Feb 2025
Adult Cardiology Elective - 1 week	McMaster University	Niagara Health - St. Catharine's Site	Jul 2024

Other Relevant Electives (2 weeks)

Emergency Medicine	University of Calgary	Calgary, Alberta	Nov 2025
Emergency Medicine	University of Ottawa	The Ottawa Hospital	Oct 2025
Emergency Medicine	McMaster University	Hamilton Health Sciences	Oct 2025
Emergency Medicine	University of Toronto	Sunnybrook Health Sciences Centre	Aug 2025
Interventional Radiology	McMaster University	Niagara Health - St. Catharine's Site	Dec 2024
Emergency Medicine	McMaster University	Niagara Health - St. Catharine's Site	Aug 2024
Cardiac Surgery	McMaster University	Hamilton General Hospital	Jul 2024
Vascular Surgery	McMaster University	Niagara Health - St. Catharine's Site	Jul 2024

RESEARCH

Selected Projects and Positions

Reducing screening burden for systematic reviews: evaluating the performance of large language models for title and abstract screening

Feb 2024 — Present

Supervisor: Dr. Stephenson Strobel | Role: Lead Investigator

- Description:**
 - Led a research team of >15 students and clinicians to screen >18000 articles across two projects.
 - Built a custom online screening platform, RapidScreen, to facilitate collaboration and data collection.
 - Found that GPT-4o-mini reduced screening burden by 72-85% with excellent specificity, NPV, and sensitivity, suggesting it may be a reliable first-line screener for reviews.
 - With this work we hope to democratize the systematic review process, making evidence-synthesis feasible for small teams and community researchers who lack the manpower and funding for this labour-intensive task.
- Deliverables:**

- Poster Presentation at the **International Conference on Emergency Medicine**. Montreal, QC. May 2025.
- Poster Presentation at Niagara Health Knowledge Institute 2025. Niagara on the Lake, ON. May 2025.
- Next Steps: **Personally awarded a \$2000 MAC RES Research Excellence Scholarship** to support detailed investigations on strengths & weaknesses of LLMs for screening, replication across scientific fields, and creation of an easy-to-use AI screening tool, with planned publication by the end of the year.

Can't Beat the Clock: Evaluation of Eleven Epigenetic Clocks for the Prediction of Mortality and Length of Stay in Critically Ill ICU Patients

Feb 2024 — Present

Supervisor: Dr. Alexandra Binnie & Dr. Jennifer Tsang | Role: Lead Investigator

- **Description:** Secondary analysis of a cohort of 144 ICU patients, evaluating the ability of DNA methylation aging clocks to predict mortality of hospital admission and length of hospital stay, using Cox proportional-hazards models, logistic regression, and Bayesian modelling. Several clocks outperformed chronological age and clinical predictors (e.g. APACHE-II score) in predicting mortality, with an AUC of up to 0.8.
- **Deliverables:**
 - Oral Presentation at **17th World Congress of Intensive & Critical Care**. Vancouver, BC. Sept 2025.
 - E-Poster at the **European Society of Intensive Care Medicine Annual Congress**. Munich, Oct 2025.
 - Poster Presentation at Niagara Health Knowledge Institute: Research Day.

Do faster-trained physicians fill the gaps? Geographic concentration of emergency medicine physicians with different postgraduate training in Ontario Canada

Dec 2025 — Present

Supervisor: Dr. Stephenson Strobel | Role: Lead Investigator

- **Description:** Using scraped physician registry data, we analyze the geospatial distribution of CCFP(EM) and FRCPC emergency physicians (and family physicians) to identify clustering of certain providers around urban centers. This has implications on healthy policy and training pathways for EM in Canada and abroad.
- **Contributions:** Created a webscraper to obtain physician data from the CPSO registry; perform geospatial data analysis and visualization; and manuscript drafting/editing.
- **Deliverables:**
 - **Publication** in *Health Policy* (Kanter-Eivin et al., 2025).
 - **Moderated Poster** at the **Canadian Association of Emergency Physicians Conference 2024**.
- Next Steps: Having expanded our data to include physicians across Canada, we hope to **examine inequities in access to primary and emergency care across Canada**, particularly in rural and Indigenous communities.

Computational Research Assistant - The Gupta Genomics Lab

Jun 2020 — May 2023

Supervisor: Dr. Radhey Gupta

- **Description:**
 - Built the AppIndels.com web-server for identification of bacterial genomes based on conserved signature indels (CSIs), reliable biochemical markers identifying common ancestry among species.
 - **Developed novel “weight” based techniques** for accounting for the varying specificity of CSIs.
 - Analyzed database of >1400 published CSIs from >90 taxa, including clinically relevant species.
 - Developed several quality-improvement and efficiency tools for use of the lab, and maintained several publicly available servers for systematic microbiology and bioinformatics, including SIGProtMIT and GLEANS.
- **Deliverables:** Published in *International Journal of Systematic and Evolutionary Microbiology* (Gupta & Kanter-Eivin, 2023).

Publications

- [1] **D. Kanter-Eivin** et al., “Do faster-trained physicians fill the gaps? Geographic concentration of emergency medicine physicians with different postgraduate training in Ontario Canada.” *Health Policy*, vol. 159, p. 105360, Sep. 2025, doi: 10.1016/j.healthpol.2025.105360.
- [2] C. Armstrong, **D. Kanter-Eivin**, M. Dowling, et al., “Accuracy of Triage Nurses in Predicting Patient Admissions: Retrospective, Large- sample Evidence from a Community Emergency Department.” *Western Journal of Emergency Medicine: Integrating Emergency Care with Population Health*, Sep. 2025, doi: 10.5811/westjem.39947.

- [3] R. S. Gupta and **D. A. Kanter-Eivin**, “AppIndels.com server: a web-based tool for the identification of known taxon-specific conserved signature indels in genome sequences. Validation of its usefulness by predicting the taxonomic affiliation of >700 unclassified strains of *Bacillus* species,” *International Journal of Systematic and Evolutionary Microbiology*, vol. 73, no. 5, 2023, doi: [10.1099/ijsem.0.005844](https://doi.org/10.1099/ijsem.0.005844).
- [4] **D. Kanter-Eivin**, H. Son, A. Steiner, and S. Strobel, “The Effects of Allowing Professional Incorporation on Physician Labour Supply,” *medRxiv*, p. 2025, Jan. 2025, doi: [10.1101/2025.03.14.25323936](https://doi.org/10.1101/2025.03.14.25323936). Under Review at *Journal of Health Economics*.
- [5] **D. Kanter-Eivin** et al., “Effects of Lithium therapy on blood glucose and insulin levels in individuals with bipolar disorder: a systematic review.” Revise & Resubmit at *Journal of Affective Disorders*. Preprint available online at: david.kantereivin.ca/LiGlu
- [6] U. Sheth, **D. Kanter-Eivin**, D. Nguyen, Y. Aldabagh, and R. Siroosi, “AI in Medical Education: A Workshop by Students, for Students.” Under Review at *Medical Science Educator*. Preprint available online at: david.kantereivin.ca/AiMedEd

LEADERSHIP & ADVOCACY

Vice-President Internal - Niagara

2023 — Present

McMaster Medical Student Council (Class of 2026)

- Represent the Niagara Campus on academic, social, and wellness within the student government, to faculty, and nationally. Advocate longitudinally for program improvement, foster community, and support peers.
 - Sit on the Niagara Campus Executive Committee, and the NRC Wellness committee.
- Overhauled MMSC processes including redesigned financial system (significantly **increasing available funding** for students), **three-year council obligations** and handover practices.
- Oversee all MMSC committees and interest groups, including defining their duties, attending key meetings, allocating funding, and supporting their initiatives.
 - Placed >120 students across three campuses into committees based on optimal matches, **creating an automated match system** successfully used for 3 years by successors (>350 students placed).
 - Created a **novel training process and centralized communication platform** for interest group chairs and committees to streamline handovers and ensure the longevity of IGs and committees.
- Passionately advocating for **Distributed Medical Education** improvements and equity across Canada, at conferences including the International Conference on Medical Education 2024 and Canadian Foundation of Medical Students AGM.
 - Successfully advocated for the **creation of the CFMS Satellite Campus Round Table**, and an inaugural member. Continue to serve to-date, providing support and sharing lessons with DME sites across Canada.
- With other VP Internals, **chaired the Intro to Medicine Committee**, organizing a welcome week and iMed month long curriculum for the Class of 2027. Key accomplishments include the **revival of the iMed “Niagara Day”** with funding for transportation and activities to showcase the region and our campus, and facilitate a sense of connectedness across campuses.
- Administered two elections securely including implementation of automatic ranked-ballot tallying.
 - Designed a weighted vote system for regional campus positions to ensure equitable elections.
- Organized the design, manufacturing, and distribution of a student-led “MacMed Patch” for the first time ever, creating a new tradition; and organized customized Patagonia merch orders for several years of Niagara learners (including the Family Medicine program).

Founding Co-Chair - Artificial Intelligence in Medicine Interest Group

2024 — Present

Michael G. DeGroote School of Medicine

- Co-organized and led multiple interprofessional sessions attended by >100 medical, nursing, PT students: “**AI in Healthcare: A Primer**”, rated >4/5 on average. Designed and led a lecture and group activity.
 - Students learned the core **professional competencies of AI in healthcare**: the basics of AI, use in different areas of healthcare, causes of bias and error, and responsible use and appraisal using the validated APPRAISE-AI tool. Lessons were reinforced with small group collaborative case-studies.

- Dissemination of session development/content, quality improvement analysis, and lessons learned are in-progress via an under review manuscript.
- Championed the **development of an artificial intelligence curriculum into UGME**, thus far resulting in a student-led professional competencies session, a transition to clerkship large-group session, and integration into the longitudinal radiology curriculum.
- A full professional competencies session, with small group components, is planned for the upcoming class.

Campus Ambassador for Niagara Regional Campus Video

2024 — 2025

McMaster UGME Communications

- Featured in the updated [NRC Campus Video](#), and [UGME Mini Documentary](#).
- Organized filming with clinical mentors and peers to showcase the vibrant Niagara campus community, incredible opportunities for learning and fun around the region, and unique relationship with faculty.

Internal Medicine Selectives Clerkship Committee Representative

Jan 2025 — Present

Michael G. DeGroote School of Medicine

- Sit on the specialty selectives committee, responsible for improving the TtR Selectives rotation and medical subspecialties, to represent Niagara campus students, and cross-campus colleagues.

NRC Wellness Committee Executive

2023 — Present

McMaster Medical Student Council (Class of 2026)

- Co-organized several wellness events including Sushi and Pizza lunches, therapy dogs, and massage day (via a collaboration with Niagara College's Massage Therapy program); assisted in budgetary management; and liaised with the Hamilton Wellness Committee Executive.

Level Representative → Senior Program Advisor → Vice-President Academic

2020 — 2023

Society of Arts & Science Students, McMaster University

- Elected four times to serve as an advocate for classmates for curriculum improvement, wellness, and constitutional reform.
- Organized many wellness events, the ArtSci Agora (a seminar between students and faculty on philosophical topics and world issues, designed to foster intellectual curiosity), and sat on the program advisory committee.

Founding Vice-President External - AI in Medicine Undergraduate Society

Sep 2021 — Apr 2023

McMaster Undergraduate Students Union

- A club for undergraduate students exploring, researching, and interested in the intersection of AI and medicine.
- Organized multiple teaching seminars, built faculty and industry contacts to create opportunities for students.

TEACHING & MENTORSHIP

Founder and Head Administrator

Jun 2017 — Present

Programmers Palace Online Programming Community

- Grew an **8000+ active member** community, tutoring 18,000+ members over 8 years to solve problems and understand challenge concepts in several programming languages such as Java, Python, R, HTML/CSS, JavaScript, C++, and Rust.
- Developed and taught a live Java curriculum to 200+ students in 14 weekly classes (free of charge), facilitated numerous coding competitions and events.

Inaugural Niagara Campus Mock OSCE Organizer

Apr 2024 — Nov 2025

- Re-established the Niagara Campus "Mock OSCE", organizing **two "Mock OSCEs"** for 32 students at the Niagara Regional Campus, including selection of clinical cases, recruitment of Standardized Patients, assessors, and volunteers; and collection of feedback for QI for subsequent MOSCEs.
- Established a third party **funding arrangement** to ensure continued funding for future mock OSCEs; and assisted in the planning of subsequent mock OSCEs.

Co-Founder, NRC Pre-Clerkship Clinical Skills Peer-Tutoring Initiative

Sep 2024 — Jan 2025

- Organized multiple practice sessions for the Class of 2027 to practice clinical skills, prepare for their OSCE, and connect with upper years.
 - Organized recruitment of volunteer SPs and assessors, selection of clinical cases, and room bookings.
 - Attended all sessions providing direct support and one-on-one mentoring.

Co-Founder, Niagara Campus Research and Clinical Talks Days Aug 2024 — Oct 2024

- Organized two half-day events for NRC students to present and receive feedback on ongoing research projects; and to share clinical pearls, approaches, and experiences from early clinical exposures.
- Developed a robust succession plan and recruited future years to ensure continuation of these events.

Teaching Assistant - COMPSCI 1MD3 (5 semesters) Jun 2020 — Dec 2022

McMaster University: Faculty of Engineering *Hamilton, ON*

- Led a team small team of TAs for a >300 person introductory programming course, as **head TA**; developed testing materials, assignments, test cases, and tutorial teaching plans, working closely with colleagues and professor for 5 semesters.
- Later, took on a primary **student support** role, providing individual help, guidance, and meetings for students struggling to succeed in the course.

RELEVANT WORK EXPERIENCE

Physician Navigator May 2022 — Aug 2023

Southlake Regional Health Centre: Emergency Department *Newmarket, ON*

- Responsible for optimization of emergency department flow, assisting emergency physicians by completing non-clinical tasks, providing patient support, and preparing relevant clinical history and documentation.
- Developed EMR quality-of-life add-on for common EMR tasks, trending and printing patient results, and detecting accidental discharges or clerical errors affecting patient care.

Audio/Visual and Events Technician 2021 — 2023

AVTEK Event Services, McMaster Students Union *Hamilton, ON*

- Supported campus events organizers in planning and executing events—from welcome week to alumnus events—around campus, while developing customer **service and leadership skills in a high stress environment**.
- Organized event technical aspects including lighting design and programming, live sound mixing, and set building. **Coordinated and communicated with large teams in fast-paced environments.**

OTHER EXTRA-CURRICULAR ACTIVITIES

Corporal of Administration & Medical First Responder 2022 — Present

St John's Ambulance - Hamilton Branch

- Responsible for managing membership information, hours, and administrative concerns, creating automated processes for most routine tasks. Overhauled recruitment and interview process, and trained new responders.
- Built a membership portal website and NFC system to automate tracking of training attendance, medical equipment use, member administration, and event sign up.
- Volunteered **≥1000 hours providing emergency medical services** for events around Hamilton, Toronto, and Niagara, including Pride Toronto, Toronto Marathon, Around the Bay, Warming Shelters, and the eclipse.
 - Received a Life Saving Award for the use of an AED and resuscitation in the field in the course of duties.

COVID-19 Vaccine Clinic Co-Coordinator 2021

Scarborough Vaccine Accessibility Clinic

- Organized two vaccine clinics, **vaccinating >150 persons with intellectual disabilities** requiring additional support.
- Coordinated clinical and non-clinical volunteers, secured a venue, and supported patients and caregivers through education, registration, and liaising with the clinical team to arrange pre-planned accommodations individualized to the patient.

Stage Manager & Technical Director

2022 — Present

MacMed Musical & ArtSci Musical

- Organized and trained technical team, led rehearsals, and devised a technical plan for 4 years of musicals.
 - Collaborated with creative teams, actors, venue staff, and stage crews to create a wonderful show!
 - More than just a technical task, theatre lighting, sound, and special effects is a creative passion of mine. Helping realize and develop a creative vision with technology is exciting; this continued over 8 years of musical theatre involvement since early high school.

PROFESSIONAL DEVELOPMENT ACTIVITIES

Certifications

Advanced Medical First Responder (AMFR), St. John Ambulance ([1485743977](#)) Nov 2022 — 2025

Incident Management System 100, for St. John Ambulance Sep 2025

Professional Memberships

Canadian Society of Internal Medicine

Canadian Association of Emergency Physicians

Canadian Critical Care Society

Society of Rural Physicians of Canada

Canadian Cardiovascular Society

College of Family Physicians of Canada

Conferences Attended (in addition to the above)

- Book to Bedside 2023. Hamilton, ON.
- Book to Bedside 2025. Hamilton, ON.
- International Conference on Academic Medicine (ICAM 2024). Vancouver, BC.
- Canadian Federation of Medical Students Annual General Meeting 2024. Moncton, NB.

Technical Skills & Languages

- **Languages:** English (native), French (conversational), Russian (casual).
- **Programming Languages:** R, Julia, Python, JavaScript, C/C++, HTML/CSS, Java, Bash, LaTeX, Typst.

HONOURS AND AWARDS

- **Niagara Regional Campus Founder's Award** (2025).
Awarded to the student from the graduating class who best exemplifies the Founders' ideals of community, scholarship, innovation, and collegiality, as chosen by the Regional Assistant Dean and nominated by 3rd year medical students at the campus.
- **MAC RES Research Excellence Scholarship** (2025).
- **NRC Niagara Campus Student Support Award – Nominee** (2024).
- **Canadian Patent Holder: "Laminate, Process, and Use"** (no. [CA2959107](#)).
Manufacturing process, product, and use for a recyclable, composable, and biodegradable coffee cup.
- **LORAN Scholarship National Semi-Finalist** (2020).
Selected based on the criteria of integrity, courage, compassion, determination, and personal autonomy.
- **Raymond Lee Award for Community Service** (2020).
Awarded for exemplary contributions to through community service and by exemplifying strong leadership.

PERSONAL INTERESTS / HOBBIES

- Love to travel: across Canada, into the wilderness (and camping), through historic cities, and hiking!
- Musical Theatre, in particular lighting, special effects, sound design, stage management, and set design.
- Cooking elaborate brunch meals—skillet eggs Benny and every type of potato you can think of!
- Board Games and Video Games, including city builders, strategy games, flight sims, and wilderness survival.